

What's so unique about working with COMs?

By definition, COMs are almost always used in a hybrid model where customers provide a carrier card that is specific to their market and application. We provide design guides and technical support to help with the carrier design. But in many cases, we collaborate during the development phase. Specifications such as COM Express contain options and, as experts, we guide our customers through the process and often provide simple BIOS changes to make their product perform as expected. It is very important that a customer chooses a COM from a vendor they can trust.

What's the next evolution of COMs?

We will see a much broader portfolio of devices being used on COMs in the future, which will broaden the use cases. Historically, COMs have typically been based on Intel x86 architecture devices. But last year, Emerson announced a range of COMs based on the Freescale QorIQ range of processors. As compute density and performance increase, more customers are adopting plug-on compute modules. This, in turn, is driving a greater range of COMs that are more application-specific.

How does Emerson ensure energy-efficiency of its products?

We take together various bits and pieces of components from the silicon community and put these together at a board level to create embedded computing products.

Emerson Network Power looks at a product using a grid-to-chip concept. The embedded products that target telecommunications or data centre servers are efficient power products that go into the server or the bottom of the ATCA cabinet. These are certified with platinum, silver or gold efficiency standards as published by Energy Star—providing more than 95 per cent efficiency on many products.

Emerson is also a board member of the Green Grid organisation, which is a global consortium comprising companies such as Intel, HP, Schneider and many other Fortune 500 companies with the aim of developing more efficient data and telecom products. We also provide some of the most efficient cooling systems to cool the cabinet and complete power management solutions that enable users to optimise the efficiency of their IT and telecom installations.

How does Emerson Embedded Computing lower the cost of ownership for its customers?

We provide open-standard building blocks, which can be very complex in many cases. We use the latest-generation computing architectures from companies such as Intel and Freescale, which have very high-speed memory interfaces and input/output. These are very complex products but we are able to have those available physically in

the market—what we call general availability—almost as soon as the silicon providers fully release their chips. This is because we have excellent relationships with these silicon providers.

We are able to make available those finished products and boards to our customers at a very good market rate, allowing them to spend their time developing their applications. So they don't have to spend time developing core compute or cooling technologies that go around the product. At the same time, they need not bother about managing the lifecycle of those products. Because we do all this for many customers, we are able to make the products available at a very competitive rate.

How do you increase the longevity of products?

All our products are based on open standards, which define the physical form factor and the connectivity of the product. This makes it very easy for the customer to source the next generation of a product in five or ten years' time with just a slot-in replacement. This enables them to execute a technology transition when they want.

In addition, we have developed a rigorous process to manage the lifecycle of our products by selecting the components for our products very carefully. We make sure that we have multiple sources for each component. So if one component reaches end-of-life, we have an alternative for that. We also have strategies to work through end-of-life component issues.

How do you deal with component end-of-life issues?

We have multiple strategies. We can buy components in a higher volume and store them. We can also develop fit, form and functional replacements for those components so that no change is needed in the software. So we try to get the best of both worlds: We enhance longevity of our products, and at the same time, we provide easier transition to new products using open standards.

What are the technical challenges when working with the new-generation silicon?

It is not with the geometry of the products, but it has more to do with the speed of the memory and fabric interfaces. Clock speeds are significantly higher than they were a few years ago, and we need to make sure that our products featuring these chips are completely reliable for the lifespan of the products. So we do what we call environmental design verification testing, where we test our products slightly outside the temperature range that the products were designed for.

We also slightly vary the clock speed and frequency of our products so that we can make sure that small variations in the silicon will not actually detract from the performance of board-level products. By using this method, we are able to make sure that that our board or product will work in five years time as it does today. ●